

CohortCharacteristics

An R package to characterise cohorts

2026-06-25

CohortCharacteristics



Website

CohortCharacteristics is on CRAN:

```
install.packages("CohortCharacteristics")
```

You can also install the development version from our GitHub repo:

```
remotes::install_github("darwin-eu/CohortCharacteristics")
```

The package documentation and vignettes are available on the website:

<https://darwin-eu.github.io/CohortCharacteristics/>

Let's get started

Let's create some cohorts that we will use during the presentation:

```
library(DiagrammeR)
library(DiagrammeRsvg)
library(rsvg)
library(grid)
library(png)

x <- c(list(anemia = 439777,
           rheumatoid_arthritis = 80809,
           osteoarthritis = 80180,
           pneumonia = 255848),
       getDrugIngredientCodes(cdm, c("heparin", "clopidogrel"),
                              nameStyle = "{concept_name}")) |>
  newCodeList()

cdm[["my_cohort"]] <- conceptCohort(cdm = cdm,
                                   conceptSet = x,
                                   name = "my_cohort")
```

Workflow

We have three types of functions:

- **summarise**: these functions produce a standardised summary of a cohort. This output is called `<summarised_result>`.
- **plot**: these functions produce plots from the `<summarised_result>` object generated by the *summarise* functions.
- **table**: these functions produce tables from the `<summarised_result>` object generated by the *summarise* functions.

Contents

- summariseCohortCount
- summariseCohortAttrition
- summariseCharacteristics
- summariseLargeScaleCharacteristics
- summariseCohortTiming
- summariseCohortOverlap

summariseCohortCount

Use `summariseCohortCount()` to get counts for your cohort. Pipe your cohort table into the function as shown below.

```
results <- cdm$my_cohort |>
  summariseCohortCount()
```

It returns cohort counts in a `summarised_result` object.

```
glimpse(results)
```

```
Rows: 12
Columns: 13
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "clopidogrel", "clopidogrel", "heparin", "heparin", "osteoarthritis"...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Number records", "Number subjects", "Number records", "Number subjects", "Number record...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count", "count", "count"...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", ...
$ estimate_value <chr> "100", "100", "357", "315", "35", "35", "2630", "2630", "52", "52", "18", "18"
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```

summariseCohortCount

Note

Yesterday we saw the function `cohortCount()` to see the counts of a `<cohort_table>`.

```
cohortCount(cdm$my_cohort)
```

```
# A tibble: 6 × 3
  cohort_definition_id number_records number_subjects
      <int>           <int>           <int>
1             1           100             100
2             2           357             315
3             3            35              35
4             4          2630            2630
5             5            52              52
6             6            18              18
```

The function `summariseCohortCount()` produces a similar result but in a standardised output (`<summarised_result>`).

tableCohortCount

To get a tidy table from the `<summarised_result>` object.

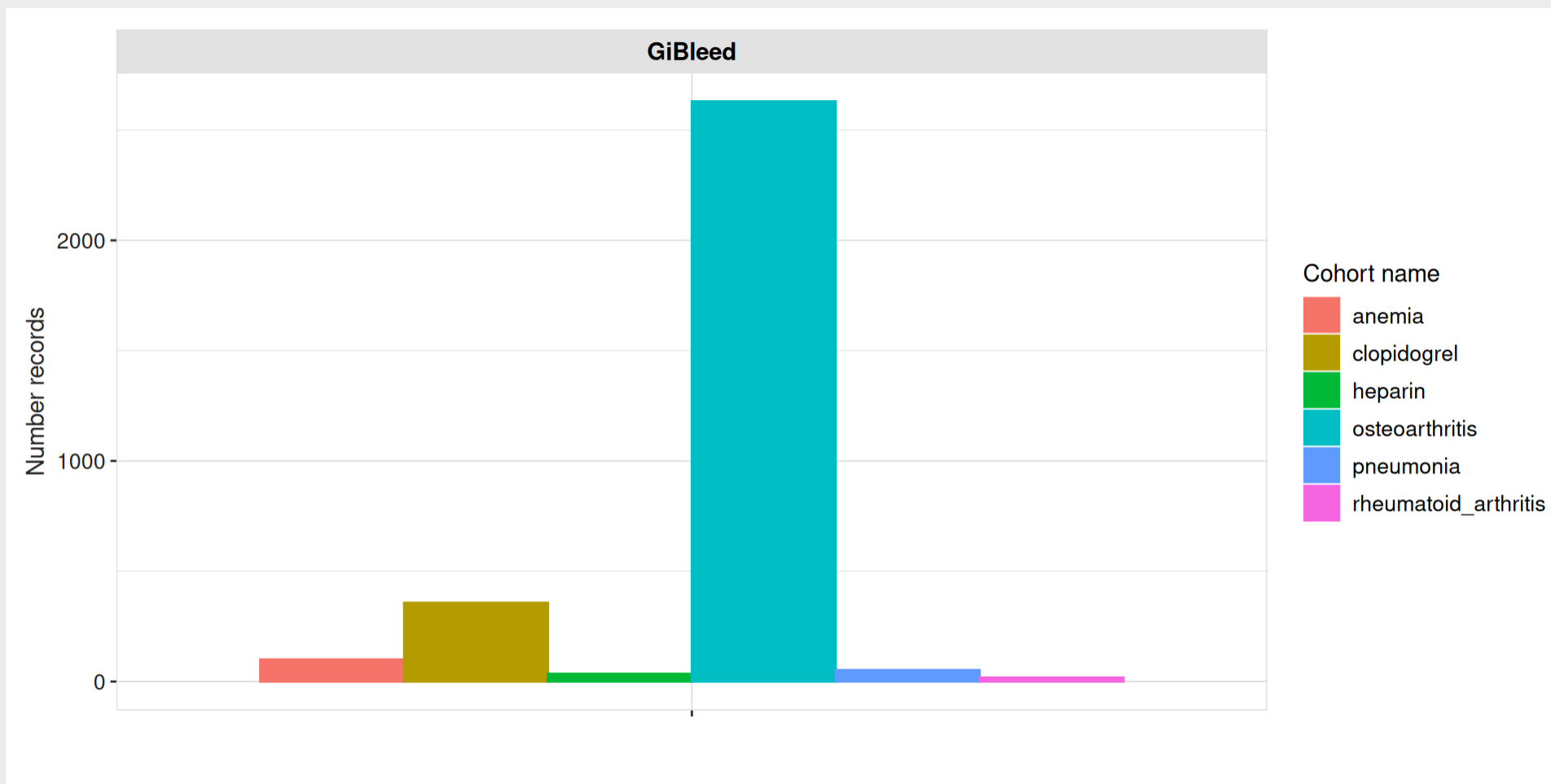
```
tableCohortCount(result = results)
```

CDM name	Variable name	Estimate name	Cohort name					
			anemia	clopidogrel	heparin	osteoarthritis	pneumonia	rheumatoid_arthritis
GiBleed	Number records	N	100	357	35	2,630	52	18
	Number subjects	N	100	315	35	2,630	52	18

plotCohortCount

Use `plotCohortCount()` to visualise cohort counts as a *ggplot2* object.

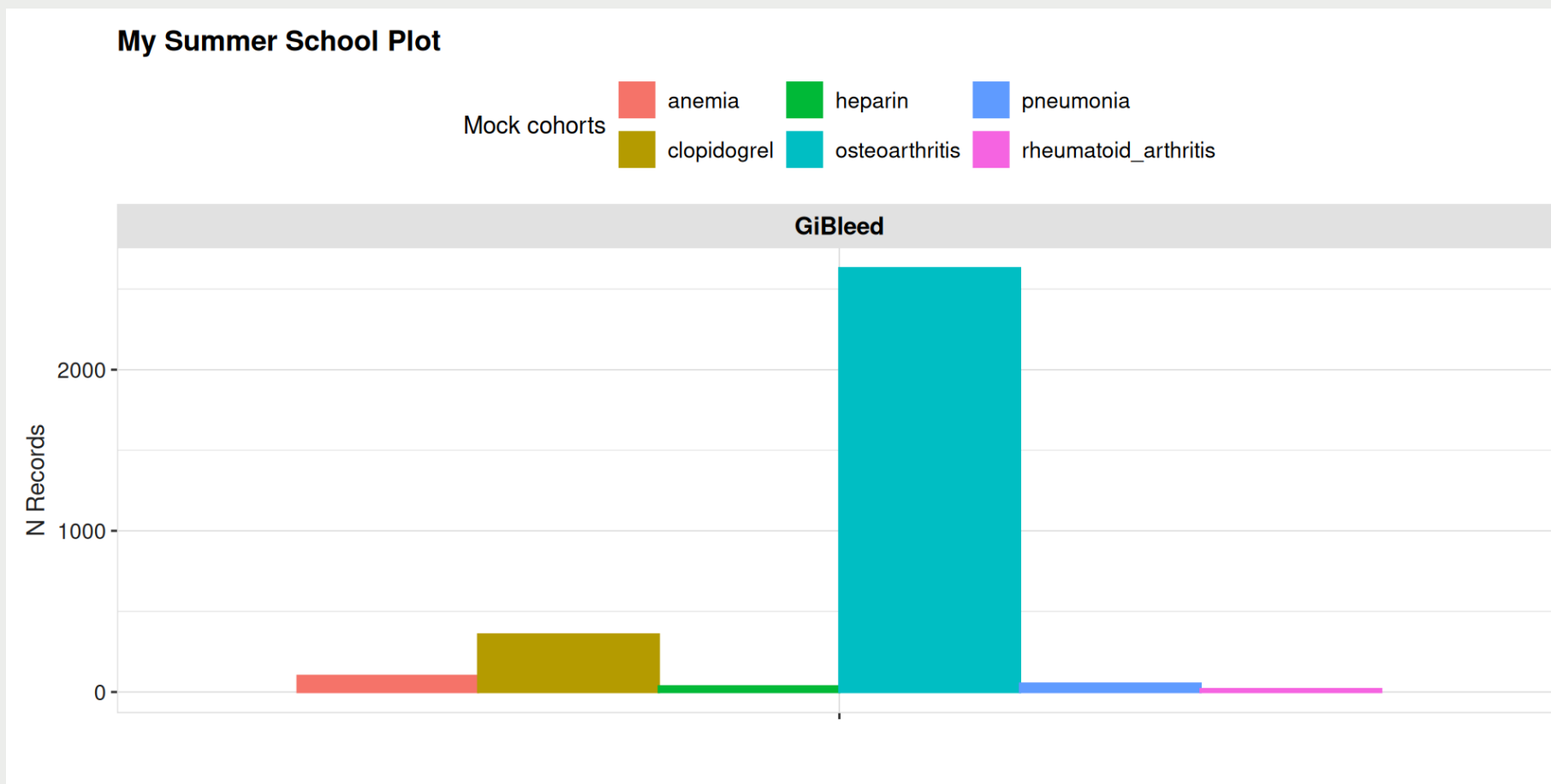
```
results |>  
  filter(variable_name == "Number records") |>  
  plotCohortCount(colour = "cohort_name")
```



plot functions in ggplot

The plotting functions return ggplot2 objects that you can edit later if you are familiar with the ggplot2 interface. For example, here we change the legend and the axis titles:

```
p <- results |>
  filter(variable_name == "Number records") |>
  plotCohortCount(colour = "cohort_name")
p +
  labs(y = "N Records", fill = "Mock cohorts", colour = "Mock cohorts", title = "My Summer School Plot") +
  theme(legend.position = "top")
```



summariseCohortAttrition

To summarise cohort attrition, pipe your cohort table into the function as shown below.

```
results <- cdm$my_cohort |>
  summariseCohortAttrition()
```

It returns cohort attrition in a **summarised_result** object.

```
glimpse(results)
```

Rows: 96

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2...
$ cdm_name      <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name    <chr> "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason"...
$ strata_level   <chr> "Initial qualifying events", "Initial qualifying events", "Initial qualifying events", "...
$ variable_name  <chr> "number_records", "number_subjects", "excluded_records", "excluded_subjects", "number_re...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ...
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count", "count", "count", "count"...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", ...
$ estimate_value <chr> "102", "102", "0", "0", "100", "100", "2", "2", "100", "100", "0", "0", "100", "100", "0...
$ additional_name <chr> "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id..."
```

summariseCohortAttrition



Note

Yesterday we saw the function `attrition()` to see the attrition of a `<cohort_table>`.

```
attrition(cdm$my_cohort)
```

```
# A tibble: 24 × 7
```

	cohort_definition_id	number_records	number_subjects	reason_id	reason	excluded_records	excluded_subjects
	<int>	<int>	<int>	<int>	<chr>	<int>	<int>
1	1	102	102	1	Initial qualif...	0	0
2	1	100	100	2	Record in obse...	2	2
3	1	100	100	3	Not missing re...	0	0
4	1	100	100	4	Merge overlapp...	0	0
5	2	363	321	1	Initial qualif...	0	0
6	2	357	315	2	Record in obse...	6	6
7	2	357	315	3	Not missing re...	0	0
8	2	357	315	4	Merge overlapp...	0	0
9	3	35	35	1	Initial qualif...	0	0
10	3	35	35	2	Record in obse...	0	0

```
# i 14 more rows
```

The function `summariseCohortAttrition()` produces a similar result but in a standardised output (`<summarised_result>`).

tableCohortAttrition

To get a tidy table from the `<summarised_result>` object.

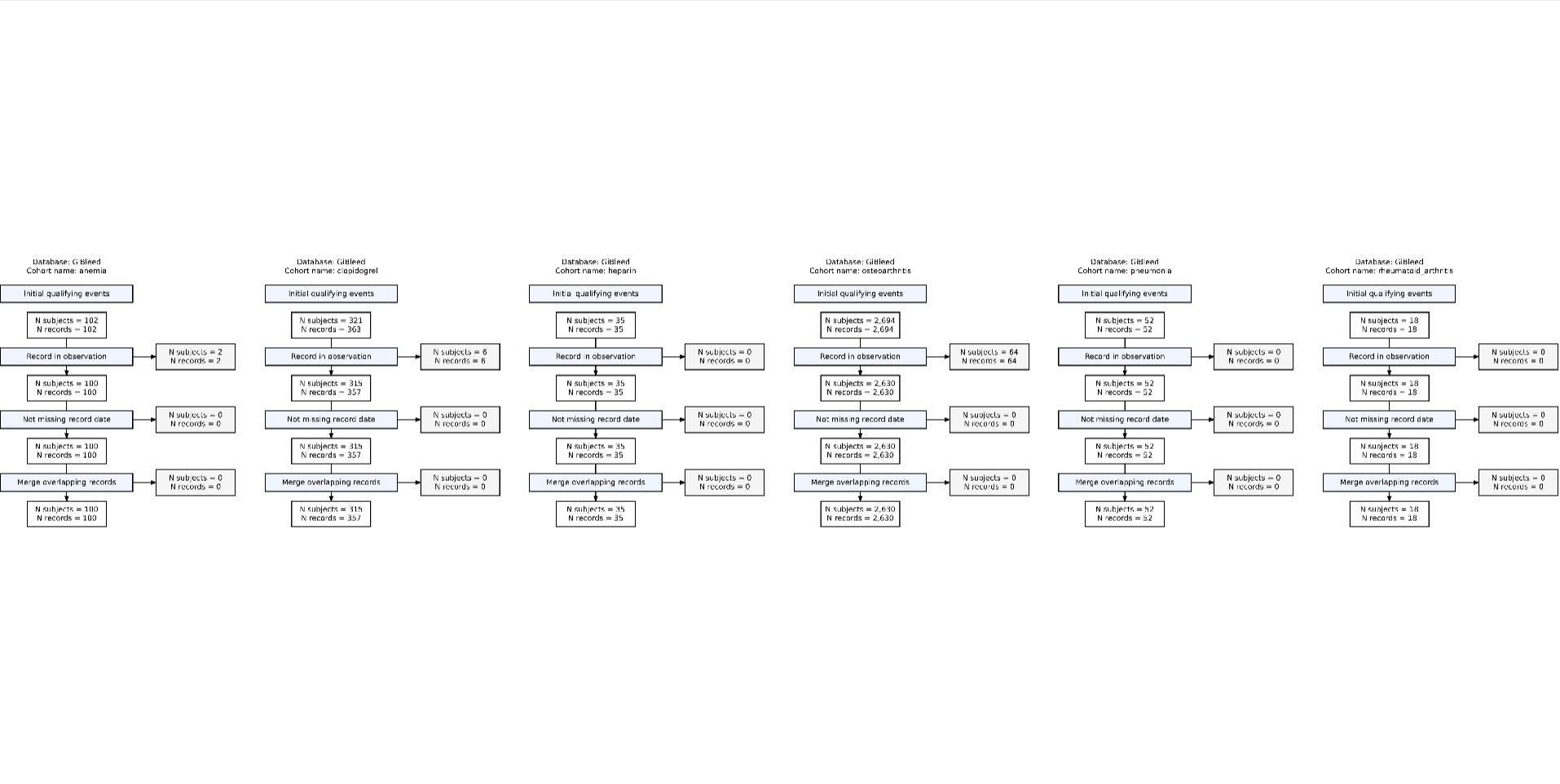
```
tableCohortAttrition(result = results)
```

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
GiBleed; anemia				
Initial qualifying events	102	102	0	0
Record in observation	100	100	2	2
Not missing record date	100	100	0	0
Merge overlapping records	100	100	0	0
GiBleed; clopidogrel				
Initial qualifying events	363	321	0	0
Record in observation	357	315	6	6
Not missing record date	357	315	0	0
Merge overlapping records	357	315	0	0
GiBleed; heparin				
Initial qualifying events	35	35	0	0
Record in observation	35	35	0	0
Not missing record date	35	35	0	0
Merge overlapping records	35	35	0	0
GiBleed; osteoarthritis				
Initial qualifying events	2,694	2,694	0	0
Record in observation	2,630	2,630	64	64

plotCohortAttrition

You can create a cohort attrition flowchart like this:

```
plotCohortAttrition(result = results, type = "png")
```



summariseCharacteristics

The main functionality of the package is to extract cohort characteristics with `summariseCharacteristics()`.

```
results <- cdm$my_cohort |>
  summariseCharacteristics()
```

It returns cohort characteristics in a `<summarised_result>` object.

```
glimpse(results)
```

```
Rows: 306
Columns: 13
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Number records", "Number subjects", "Cohort start date", "Cohort start date", "Cohort s...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Female", "F...
$ estimate_name  <chr> "count", "count", "min", "q25", "median", "q75", "max", "min", "q25", "median", "q75", "...
$ estimate_type  <chr> "integer", "integer", "date", "date", "date", "date", "date", "date", "date", "date", "date", "d...
$ estimate_value <chr> "100", "100", "1912-04-30", "1974-01-30", "1990-12-04", "2005-08-09", "2018-11-08", "191...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```


tableCharacteristics

To get a tidy table from the `<summarised_result>` object.

```
tableCharacteristics(result = results, header = "cohort_name")
```

CDM name	Variable name	Variable level	Estimate name	Cohort name					
				anemia	clopidogrel	heparin	osteoarthritis	pneumonia	rheumatoid_arthritis
GiBleed	Number records	–	N	100	357	35	2,630	52	18
	Number subjects	–	N	100	315	35	2,630	52	18
	Cohort start date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	2008-01-18 [1999-12-17 - 2014-09-06]	1978-05-19 [1968-02-18 - 1996-08-26]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2000-08-10 [1992-01-18 - 2008-01-01]
			Range	1912-04-30 to 2018-11-08	1997-01-08 to 2019-06-26	1939-05-23 to 2018-07-21	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Cohort end date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	2010-01-19 [2003-08-10 - 2015-05-14]	1978-05-19 [1968-02-18 - 1996-08-26]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2004-11-17 [1993-03-05 - 2011-05-22]
			Range	1912-04-30 to 2018-11-08	1997-01-08 to 2019-06-26	1939-05-23 to 2018-07-21	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Age	–	Median [Q25 - Q75]	32 [18 - 50]	61 [49 - 77]	29 [10 - 52]	38 [36 - 41]	82 [78 - 86]	41 [34 - 55]
			Mean (SD)	33.49 (22.14)	61.45 (18.53)	30.66 (23.98)	38.15 (3.27)	80.33 (9.48)	45.22 (15.11)
			Range	0 to 81	17 to 107	0 to 77	31 to 46	53 to 96	24 to 77
	Sex	Female	N (%)	78 (78.00%)	131 (36.69%)	29 (82.86%)	1,341 (50.99%)	36 (69.23%)	8 (44.44%)
		Male	N (%)	22 (22.00%)	226 (63.31%)	6 (17.14%)	1,289 (49.01%)	16 (30.77%)	10 (55.56%)
	Prior observation	–	Median [Q25 - Q75]	12,045 [6,843 - 18,706]	22,344 [17,955 - 28,280]	10,952 [3,694 - 19,346]	14,097 [13,161 - 15,039]	30,079 [28,538 - 31,726]	15,330 [12,866 - 20,440]
			Mean (SD)	12,570.42 (8,107.27)	22,612.10 (6,778.54)	11,400.40 (8,893.69)	14,105.97 (1,193.14)	29,498.13 (3,467.33)	16,870.94 (5,513.54)
			Range	0 to 29,930	6,257 to 39,424	46 to 28,472	11,340 to 17,044	19,485 to 35,210	9,125 to 28,469
	Future observation	–	Median [Q25 - Q75]	9,642 [4,596 - 14,788]	2,268 [847 - 5,040]	13,406 [7,882 - 17,001]	6,790 [3,568 - 10,791]	12 [10 - 15]	6,066 [4,058 - 9,716]

Commonly used arguments for summariseCharacteristics

- **strata**: creates stratified results.
- **cohortId**: filters by cohort ID.
- **ageGroup**: defines the age groups to summarise.
- **tableIntersect**: count/flag/date/days
- **cohortIntersect**: count/flag/date/days
- **conceptIntersect**: count/flag/date/days
- **link to documentation**

ageGroup example

```
result <- summariseCharacteristics(cdm$my_cohort, ageGroup = list(c(0,10), c(11,18), c(19,150)))  
tableCharacteristics(result = result, header = "cohort_name")
```

ageGroup example

CDM name	Variable name	Variable level	Estimate name	Cohort name					
				anemia	clopidogrel	heparin	osteoarthritis	pneumonia	rheumatoid_arthritis
GiBleed	Number records	—	N	100	357	35	2,630	52	18
	Number subjects	—	N	100	315	35	2,630	52	18
	Cohort start date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	2008-01-18 [1999-12-17 - 2014-09-06]	1978-05-19 [1968-02-18 - 1996-08-26]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2000-08-10 [1992-01-18 - 2008-01-01]
			Range	1912-04-30 to 2018-11-08	1997-01-08 to 2019-06-26	1939-05-23 to 2018-07-21	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Cohort end date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	2010-01-19 [2003-08-10 - 2015-05-14]	1978-05-19 [1968-02-18 - 1996-08-26]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2004-11-17 [1993-03-05 - 2011-05-22]
			Range	1912-04-30 to 2018-11-08	1997-01-08 to 2019-06-26	1939-05-23 to 2018-07-21	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Age	—	Median [Q25 - Q75]	32 [18 - 50]	61 [49 - 77]	29 [10 - 52]	38 [36 - 41]	82 [78 - 86]	41 [34 - 55]
			Mean (SD)	33.49 (22.14)	61.45 (18.53)	30.66 (23.98)	38.15 (3.27)	80.33 (9.48)	45.22 (15.11)
			Range	0 to 81	17 to 107	0 to 77	31 to 46	53 to 96	24 to 77
	Age group	0 to 10	N (%)	20 (20.00%)	0 (0.00%)	9 (25.71%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
		11 to 18	N (%)	7 (7.00%)	1 (0.28%)	8 (22.86%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
		19 to 150	N (%)	73 (73.00%)	356 (99.72%)	18 (51.43%)	2,630 (100.00%)	52 (100.00%)	18 (100.00%)
	Sex	Female	N (%)	78 (78.00%)	131 (36.69%)	29 (82.86%)	1,341 (50.99%)	36 (69.23%)	8 (44.44%)
		Male	N (%)	22 (22.00%)	226 (63.31%)	6 (17.14%)	1,289 (49.01%)	16 (30.77%)	10 (55.56%)
	Prior observation	—	Median [Q25 - Q75]	12,045 [6,843 - 18,706]	22,344 [17,955 - 28,280]	10,952 [3,694 - 19,346]	14,097 [13,161 - 15,039]	30,079 [28,538 - 31,726]	15,330 [12,866 - 20,440]
			Mean (SD)	12,570.42 (8,107.27)	22,612.10 (6,778.54)	11,400.40 (8,893.69)	14,105.97 (1,193.14)	29,498.13 (3,467.33)	16,870.94 (5,513.54)

Strata example

```
result <- cdm$my_cohort |>
  addAge(ageGroup = list(c(0,10),c(11,18), c(19,150))) |>
  summariseCharacteristics(cohortId = 1,
                          strata = list("age_group"))
tableCharacteristics(result = result, header = "age_group")
```

Strata example

CDM name	Cohort name	Variable name	Variable level	Estimate name	Age group			
					overall	0 to 10	11 to 18	19 to 150
GiBleed	anemia	Number records	—	N	100	20	7	73
		Number subjects	—	N	100	20	7	73
		Cohort start date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1963-03-26 [1956-03-03 - 1972-12-27]	1969-08-19 [1963-03-03 - 1975-04-11]	1999-03-30 [1987-11-03 - 2007-06-08]
				Range	1912-04-30 to 2018-11-08	1912-04-30 to 1987-01-08	1951-04-05 to 1986-04-07	1939-01-29 to 2018-11-08
		Cohort end date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1963-03-26 [1956-03-03 - 1972-12-27]	1969-08-19 [1963-03-03 - 1975-04-11]	1999-03-30 [1987-11-03 - 2007-06-08]
				Range	1912-04-30 to 2018-11-08	1912-04-30 to 1987-01-08	1951-04-05 to 1986-04-07	1939-01-29 to 2018-11-08
		Age	—	Median [Q25 - Q75]	32 [18 - 50]	2 [1 - 5]	17 [14 - 18]	43 [28 - 56]
				Mean (SD)	33.49 (22.14)	3.30 (3.15)	15.86 (2.27)	43.45 (16.93)
				Range	0 to 81	0 to 9	12 to 18	19 to 81
		Sex	Female	N (%)	78 (78.00%)	10 (50.00%)	6 (85.71%)	62 (84.93%)
			Male	N (%)	22 (22.00%)	10 (50.00%)	1 (14.29%)	11 (15.07%)
		Prior observation	—	Median [Q25 - Q75]	12,045 [6,843 - 18,706]	1,094 [365 - 2,280]	6,570 [5,657 - 6,752]	16,060 [10,585 - 20,804]
				Mean (SD)	12,570.42 (8,107.27)	1,478.00 (1,221.91)	6,152.57 (827.72)	16,224.85 (6,180.75)
				Range	0 to 29,930	0 to 3,650	4,745 to 6,935	7,300 to 29,930
		Future observation	—	Median [Q25 - Q75]	9,642 [4,596 - 15,796]	19,970 [14,251 - 22,878]	17,974 [16,044 - 20,252]	6,808 [3,994 - 10,939]
				Mean (SD)	10,773.47 (7,880.91)	20,042.95 (6,677.93)	18,200.43 (4,119.31)	7,521.71 (5,680.34)

Intersection with another table

You can get count/flag/date/days summaries with other tables.

```
result <- cdm$my_cohort |>
  summariseCharacteristics(
    cohortId = "anemia",
    tableIntersectCount = list(
      "Number of visits prior year" = list(
        tableName = "visit_occurrence",
        window = c(-365, 0)
      )
    ),
    cohortIntersectFlag = list(
      "Previous rheumatoid arthritis diagnosis" = list(
        targetCohortTable = "my_cohort",
        targetCohortId = "rheumatoid_arthritis",
        window = c(-Inf, 0)
      )
    ),
    conceptIntersectDays = list(
      "Time since last X-ray" = list(
```

Intersection with another table

You can get count/flag/date/days summaries with other tables.

CDM name	Variable name	Variable level	Estimate name	Cohort name					
				anemia	clopidogrel	heparin	osteoarthritis	pneumonia	rheumatoid_arthritis
GiBleed	Number records	–	N	100	357	35	2,630	52	18
	Number subjects	–	N	100	315	35	2,630	52	18
	Cohort start date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	2008-01-18 [1999-12-17 - 2014-09-06]	1978-05-19 [1968-02-18 - 1996-08-26]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2000-08-10 [1992-01-18 - 2008-01-01]
			Range	1912-04-30 to 2018-11-08	1997-01-08 to 2019-06-26	1939-05-23 to 2018-07-21	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Cohort end date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	2010-01-19 [2003-08-10 - 2015-05-14]	1978-05-19 [1968-02-18 - 1996-08-26]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2004-11-17 [1993-03-05 - 2011-05-22]
			Range	1912-04-30 to 2018-11-08	1997-01-08 to 2019-06-26	1939-05-23 to 2018-07-21	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Age	–	Median [Q25 - Q75]	32 [18 - 50]	61 [49 - 77]	29 [10 - 52]	38 [36 - 41]	82 [78 - 86]	41 [34 - 55]
			Mean (SD)	33.49 (22.14)	61.45 (18.53)	30.66 (23.98)	38.15 (3.27)	80.33 (9.48)	45.22 (15.11)
			Range	0 to 81	17 to 107	0 to 77	31 to 46	53 to 96	24 to 77
	Sex	Female	N (%)	78 (78.00%)	131 (36.69%)	29 (82.86%)	1,341 (50.99%)	36 (69.23%)	8 (44.44%)
		Male	N (%)	22 (22.00%)	226 (63.31%)	6 (17.14%)	1,289 (49.01%)	16 (30.77%)	10 (55.56%)
	Prior observation	–	Median [Q25 - Q75]	12,045 [6,843 - 18,706]	22,344 [17,955 - 28,280]	10,952 [3,694 - 19,346]	14,097 [13,161 - 15,039]	30,079 [28,538 - 31,726]	15,330 [12,866 - 20,440]
			Mean (SD)	12,570.42 (8,107.27)	22,612.10 (6,778.54)	11,400.40 (8,893.69)	14,105.97 (1,193.14)	29,498.13 (3,467.33)	16,870.94 (5,513.54)
			Range	0 to 29,930	6,257 to 39,424	46 to 28,472	11,340 to 17,044	19,485 to 35,210	9,125 to 28,469
	Future observation	–	Median [Q25 - Q75]	9,642 [4,596 - 15,796]	2,268 [847 - 5,040]	13,406 [7,882 - 17,991]	6,790 [3,568 - 10,781]	12 [10 - 15]	6,066 [4,058 - 9,716]
			Mean (SD)	10,773.47 (8,084.03)	2,684.03 (5,106.17)	13,156.29 (8,884.03)	7,686.76 (5,264.20)	12.25 (2.16)	6,078.83 (4,735.70)

Intersections

Note

The arguments of the intersection functions match the *PatientProfiles* functions that we have just seen:

- `tableIntersectFlag -> PatientProfiles::addTableIntersectFlag()`
- `tableIntersectCount -> PatientProfiles::addTableIntersectCount()`
- `tableIntersectDays -> PatientProfiles::addTableIntersectDays()`
- `tableIntersectDate -> PatientProfiles::addTableIntersectDate()`
- `conceptIntersectFlag -> PatientProfiles::addConceptIntersectFlag()`
- `conceptIntersectCount -> PatientProfiles::addConceptIntersectCount()`
- `conceptIntersectDays -> PatientProfiles::addConceptIntersectDays()`
- `conceptIntersectDate -> PatientProfiles::addConceptIntersectDate()`
- `cohortIntersectFlag -> PatientProfiles::addCohortIntersectFlag()`
- `cohortIntersectCount -> PatientProfiles::addCohortIntersectCount()`
- `cohortIntersectDays -> PatientProfiles::addCohortIntersectDays()`
- `cohortIntersectDate -> PatientProfiles::addCohortIntersectDate()`

Your turn

Can you characterise the **osteoarthritis** cohort?

- Start with only demographics
- Add if they had any heparin prescription during the last 180 days (including index date).
- Add if they had any pneumonia diagnosis last 2 years (excluding index date).

►  Click to see solution

CDM name	Variable name	Variable level	Estimate name	Cohort name
				osteoarthritis
delphi	Number records	—	N	212
	Number subjects	—	N	212
	Cohort start date	—	Median [Q25 - Q75]	1993-07-01 [1976-02-12 - 1997-09-29]
			Range	1945-05-13 to 2004-12-14
	Cohort end date	—	Median [Q25 - Q75]	1993-07-01 [1976-02-12 - 1997-09-29]
			Range	1945-05-13 to 2004-12-14
	Age	—	Median [Q25 - Q75]	18 [18 - 18]
			Mean (SD)	18.00 (0.00)
			Range	18 to 18

CDM name	Variable name	Variable level	Estimate name	Cohort name
				osteoarthritis
	Sex	Female	N (%)	38 (17.92%)
		Male	N (%)	174 (82.08%)
	Prior observation	–	Median [Q25 - Q75]	5,611 [387 - 6,574]
			Mean (SD)	4,105.32 (2,777.03)
			Range	0 to 6,575
	Future observation	–	Median [Q25 - Q75]	6,603 [5,101 - 13,022]
			Mean (SD)	8,950.94 (5,459.37)
			Range	2,472 to 24,272

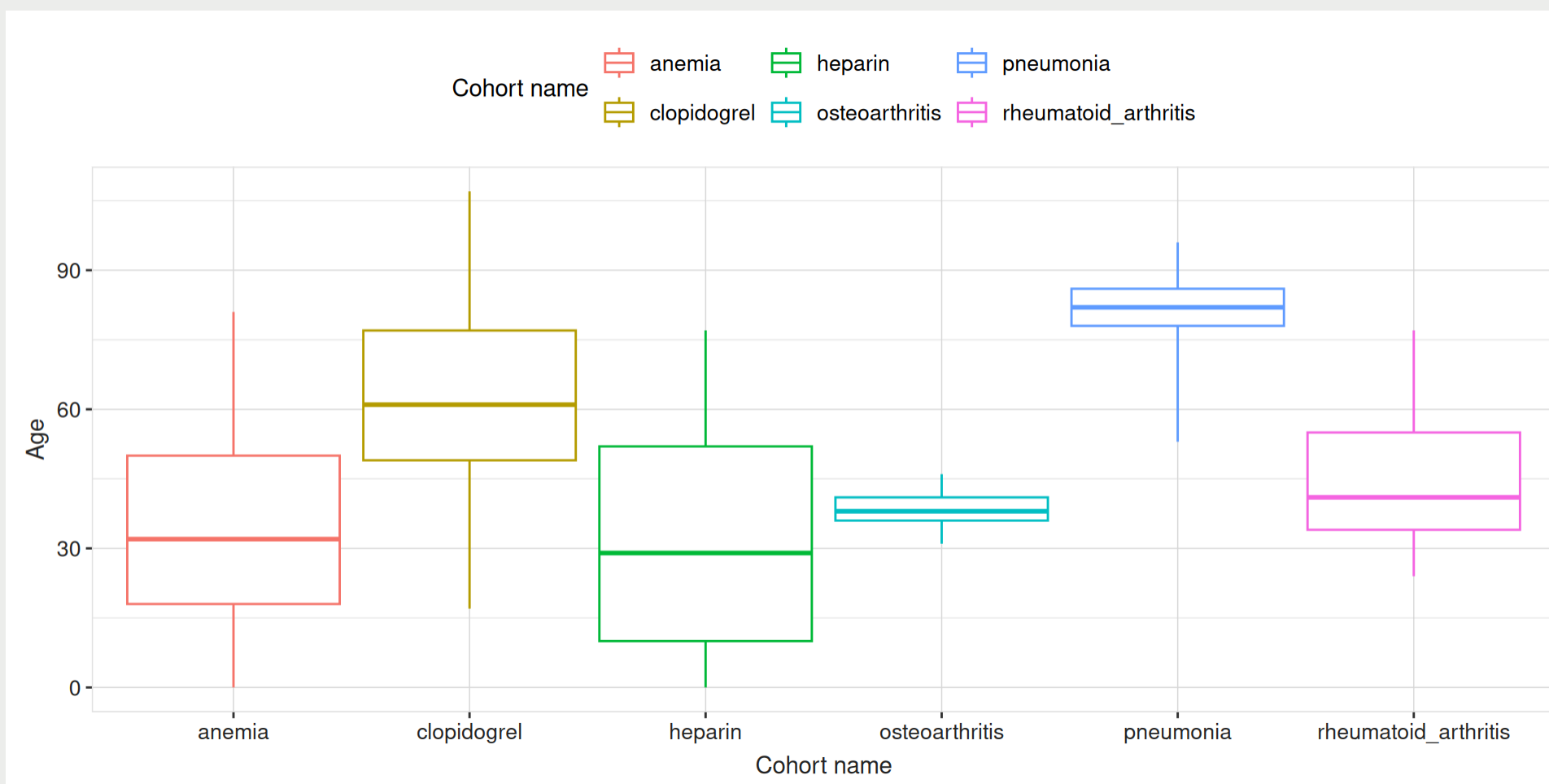
[← Back to Presentations](#)

plotCharacteristics()

You can plot a variable from the `summariseCharacteristics()` output using `plotCharacteristics()`.

```
result <- cdm$my_cohort |>
  summariseCharacteristics() |>
  filter(variable_name == "Age")

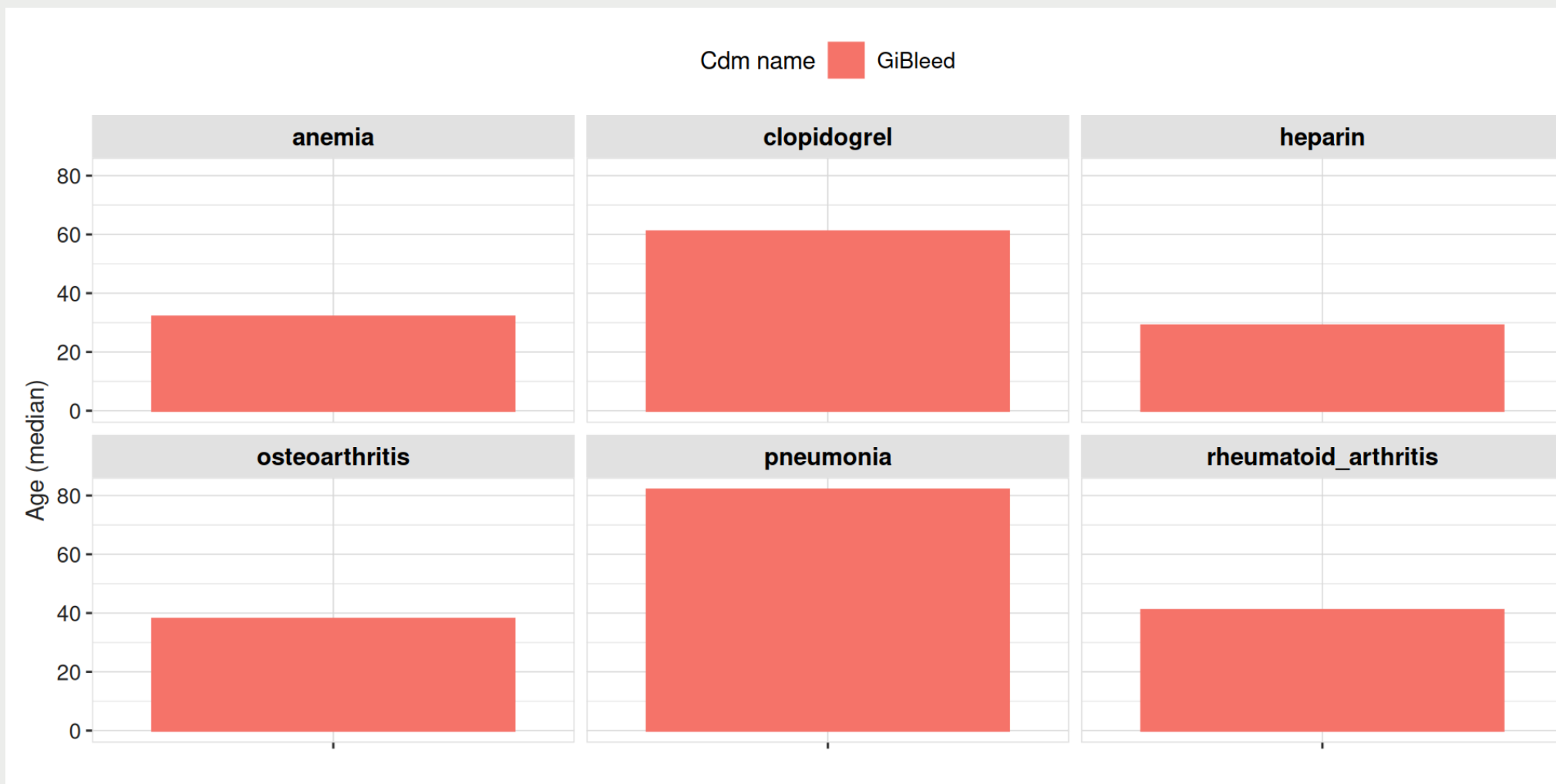
plotCharacteristics(result = result, plotStyle = "boxplot", colour = "cohort_name")
```



plotCharacteristics()

```
result <- cdm$my_cohort |>
  summariseCharacteristics() |>
  filter(variable_name == "Age" & estimate_name == "median")

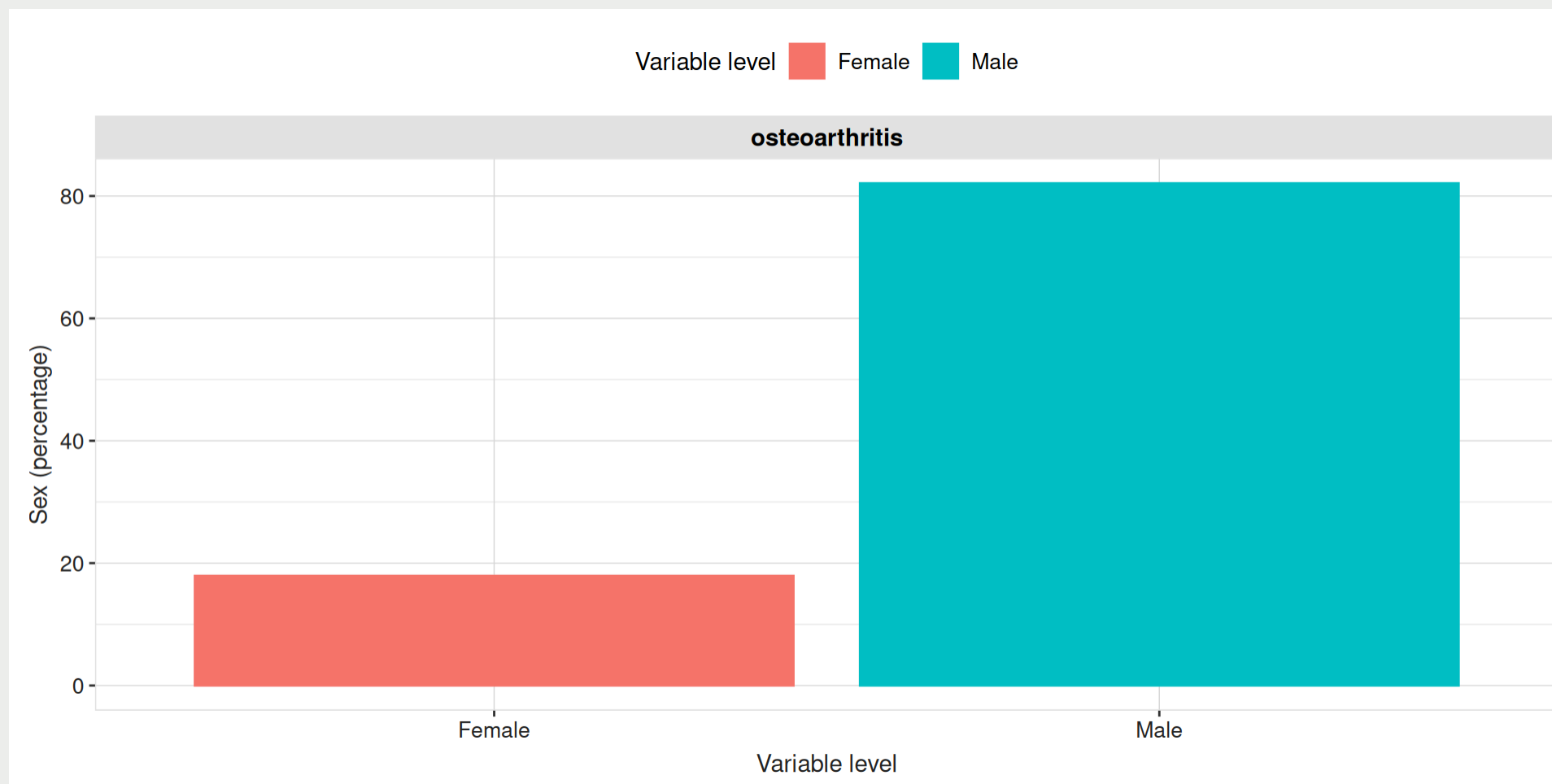
plotCharacteristics(result = result,
  plotStyle = "barplot",
  facet = "cohort_name",
  colour = "cdm_name")
```



Your turn

Can you create a bar plot the percentage of sex in the **osteoarthritis** table?

►  Click to see solution



summariseLargeScaleCharacteristics

Sometimes we might want to summarise all clinical events for cohorts across different time windows. We can do this with `summariseLargeScaleCharacteristics()`.

```
result <- cdm$my_cohort |>
  summariseLargeScaleCharacteristics(
    window = list(c(-Inf, -1), c(1, Inf)),
    eventInWindow = "condition_occurrence",
    minimumFrequency = 0.05
  )
```

```
glimpse(result)
```

Rows: 372

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"..
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Acute allergic reaction", "Acute allergic reaction", "Acute bronchitis", "Acute bronchi...
$ variable_level <chr> "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-in...
$ estimate_name  <chr> "count", "percentage", "count", "percentage", "count", "percentage", "count", "percentag...
$ estimate_type  <chr> "integer", "percentage", "integer", "percentage", "integer", "percentage", "integer", "p...
$ estimate_value <chr> "5", "5.10", "77", "78.57", "75", "76.53", "7", "7.14", "6", "6.12", "6", "6.12", "7", "...
$ additional_name <chr> "concept_id", "concept_id", "concept_id", "concept_id", "concept_id", "concept_id", "con..."
```

tableTopLargeScaleCharacteristics

`tableTopLargeScaleCharacteristics()` creates a tidy table of the most frequent concepts.

```
tableTopLargeScaleCharacteristics(result = result, topConcepts = 5)
```

	Cohort name										
Top	anemia		clopidogrel		heparin		osteoarthritis		pneumonia	rheumatoid_arthritis	
	Window										
	-inf to -1	1 to inf	-inf to -1	1 to inf	-inf to -1	1 to inf	-inf to -1	1 to inf	-inf to -1	-inf to -1	1 to inf
1	Viral sinusitis (40481087) 82 (83.7%)	Viral sinusitis (40481087) 82 (82.0%)	Viral sinusitis (40481087) 355 (99.4%)	Viral sinusitis (40481087) 186 (52.2%)	Otitis media (372328) 28 (80.0%)	Viral sinusitis (40481087) 31 (88.6%)	Viral sinusitis (40481087) 2594 (98.6%)	Viral sinusitis (40481087) 2066 (78.6%)	Alzheimer's disease (378419) 43 (82.7%)	Acute viral pharyngitis (4112343) 17 (94.4%)	Viral sinusitis (40481087) 13 (72.2%)
2	Acute bronchitis (260139) 77 (78.6%)	Acute viral pharyngitis (4112343) 70 (70.0%)	Acute viral pharyngitis (4112343) 343 (96.1%)	Streptococcal sore throat (28060) 32 (9.0%)	Acute viral pharyngitis (4112343) 25 (71.4%)	Acute viral pharyngitis (4112343) 28 (80.0%)	Acute viral pharyngitis (4112343) 2403 (91.4%)	Acute viral pharyngitis (4112343) 1664 (63.3%)	Streptococcal sore throat (28060) 43 (82.7%)	Viral sinusitis (40481087) 17 (94.4%)	Acute viral pharyngitis (4112343) 12 (66.7%)
3	Acute viral pharyngitis (4112343) 75 (76.5%)	Acute bronchitis (260139) 65 (65.0%)	Acute bronchitis (260139) 332 (93.0%)	Sinusitis (4283893) 26 (7.3%)	Viral sinusitis (40481087) 24 (68.6%)	Acute bronchitis (260139) 27 (77.1%)	Acute bronchitis (260139) 2292 (87.2%)	Acute bronchitis (260139) 1515 (57.6%)	Otitis media (372328) 41 (78.8%)	Otitis media (372328) 13 (72.2%)	Acute cholecystitis (198809) 1 (5.6%)
4	Otitis media (372328) 69 (70.4%)	Osteoarthritis (80180) 58 (58.0%)	Osteoarthritis (80180) 315 (88.2%)	Sprain of ankle (81151) 24 (6.7%)	Acute bronchitis (260139) 20 (57.1%)	Osteoarthritis (80180) 24 (68.6%)	Otitis media (372328) 1935 (73.6%)	Chronic sinusitis (257012) 251 (9.5%)	Sprain of ankle (81151) 35 (67.3%)	Osteoarthritis (80180) 12 (66.7%)	Alzheimer's disease (378419) 1 (5.6%)
5	Streptococcal sore throat (28060) 59 (60.2%)	Angiodysplasia of stomach (4310024) 9 (9.0%)	Otitis media (372328) 262 (73.4%)	Acute bacterial sinusitis (4294548) 21 (5.9%)	Cystitis (195588) 3 (8.6%)	Angiodysplasia of stomach (4310024) 3 (8.6%)	Streptococcal sore throat (28060) 1399 (53.2%)	Cerebrovascular accident (381316) 194 (7.4%)	Laceration of foot (4109685) 5 (9.6%)	Streptococcal sore throat (28060) 9 (50.0%)	Anemia (439777) 1 (5.6%)

tidy

You can create a tidy table that you can explore with the `tidy()` function:

```
tidy(result)
```

```
# A tibble: 186 × 10
```

	cdm_name	cohort_name	variable_name	variable_level	concept_id	count	percentage	analysis	table_name	type
	<chr>	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>
1	GiBleed	anemia	Acute allergic reacti...	-inf to -1	4084167	5	5.1	standard	condition...	event
2	GiBleed	anemia	Acute bronchitis	-inf to -1	260139	77	78.6	standard	condition...	event
3	GiBleed	anemia	Acute viral pharyngit...	-inf to -1	4112343	75	76.5	standard	condition...	event
4	GiBleed	anemia	Angiodysplasia of sto...	-inf to -1	4310024	7	7.14	standard	condition...	event
5	GiBleed	anemia	Appendicitis	-inf to -1	440448	6	6.12	standard	condition...	event
6	GiBleed	anemia	Child attention defici...	-inf to -1	440086	6	6.12	standard	condition...	event
7	GiBleed	anemia	Concussion with loss ...	-inf to -1	375671	7	7.14	standard	condition...	event
8	GiBleed	anemia	Cystitis	-inf to -1	195588	7	7.14	standard	condition...	event
9	GiBleed	anemia	Facial laceration	-inf to -1	4156265	9	9.18	standard	condition...	event
10	GiBleed	anemia	First degree burn	-inf to -1	4296204	8	8.16	standard	condition...	event

```
# i 176 more rows
```

tidy

Note

The `tidy()` function works for any `<summarised_result>` object that we create.

summariseCohortOverlap

When creating multiple cohorts, we might be interested in the overlap of individuals between those cohorts. `summariseCohortOverlap()` does this for you.

```
result <- cdm$my_cohort |>
  summariseCohortOverlap()
```

```
glimpse(result)
```

Rows: 180

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name_reference &&& cohort_name_comparator", "cohort_name_reference &&& cohort_na...
$ group_level    <chr> "anemia &&& clopidogrel", "anemia &&& clopidogrel", "anemia &&& clopidogrel", "anemia &&...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Only in reference cohort", "In both cohorts", "Only in comparator cohort", "Only in ref...
$ variable_level <chr> "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Sub...
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count", "count", "count", "count"...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", ...
$ estimate_value <chr> "92", "8", "307", "99", "1", "34", "3", "97", "2533", "93", "7", "45", "99", "1", "17", ...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```

tableCohortOverlap

You can also get a table with `tableCohortOverlap()`.

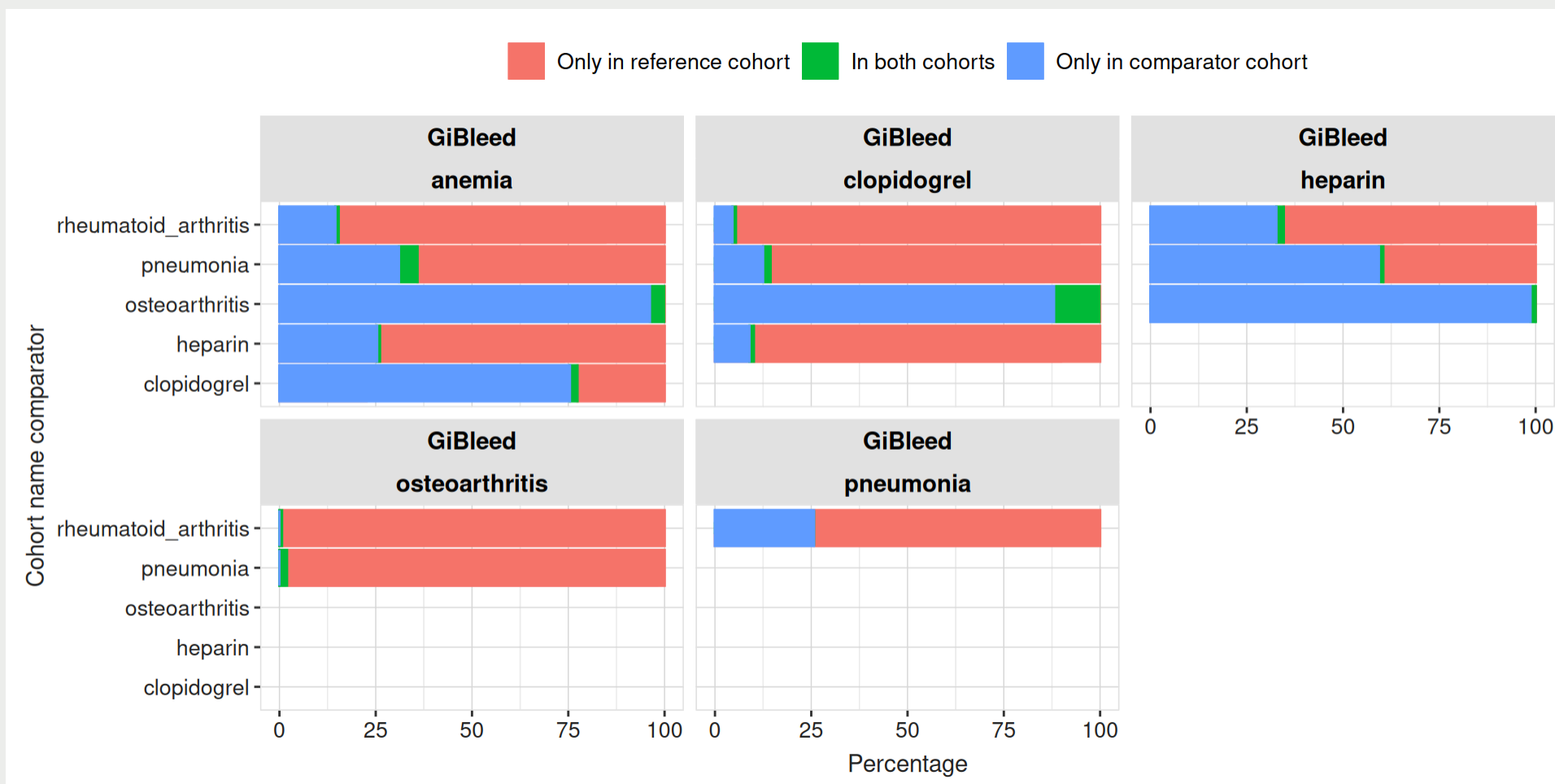
```
tableCohortOverlap(result)
```

Cohort name reference	Cohort name comparator	Estimate name	Variable name		
			Only in reference cohort	In both cohorts	Only in comparator cohort
GiBleed					
anemia	clopidogrel	N (%)	92 (22.60%)	8 (1.97%)	307 (75.43%)
	heparin	N (%)	99 (73.88%)	1 (0.75%)	34 (25.37%)
	osteoarthritis	N (%)	3 (0.11%)	97 (3.68%)	2,533 (96.20%)
	pneumonia	N (%)	93 (64.14%)	7 (4.83%)	45 (31.03%)
	rheumatoid_arthritis	N (%)	99 (84.62%)	1 (0.85%)	17 (14.53%)
clopidogrel	heparin	N (%)	311 (89.88%)	4 (1.16%)	31 (8.96%)
	osteoarthritis	N (%)	5 (0.19%)	310 (11.76%)	2,320 (88.05%)
	pneumonia	N (%)	308 (85.56%)	7 (1.94%)	45 (12.50%)
	rheumatoid_arthritis	N (%)	312 (94.55%)	3 (0.91%)	15 (4.55%)
heparin	osteoarthritis	N (%)	0 (0.00%)	35 (1.33%)	2,595 (98.67%)
	pneumonia	N (%)	34 (39.53%)	1 (1.16%)	51 (59.30%)
	rheumatoid_arthritis	N (%)	34 (65.38%)	1 (1.92%)	17 (32.69%)
osteoarthritis	pneumonia	N (%)	2,578 (98.02%)	52 (1.98%)	0 (0.00%)
	rheumatoid_arthritis	N (%)	2,612 (99.32%)	18 (0.68%)	0 (0.00%)
pneumonia	rheumatoid_arthritis	N (%)	52 (74.29%)	0 (0.00%)	18 (25.71%)

plotCohortOverlap

You can also get a plot with `plotCohortOverlap()`.

```
plotCohortOverlap(result = result, uniqueCombinations = TRUE)
```



summariseCohortTiming

When creating multiple cohorts, we might be interested in the timing of entry between cohorts. `summariseCohortTiming()` does this for you.

```
result <- cdm$my_cohort |>
  summariseCohortTiming(restrictToFirstEntry = TRUE)

glimpse(result)
```

Rows: 20,724

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name_reference &&& cohort_name_comparator", "cohort_name_reference &&& cohort_na...
$ group_level    <chr> "anemia &&& clopidogrel", "anemia &&& clopidogrel", "anemia &&& clopidogrel", "anemia &&...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "number records", "number subjects", "days_between_cohort_entries", "days_between_cohort...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, "density_001", "density_002", "density_003", "density_004", ...
$ estimate_name  <chr> "count", "count", "min", "q25", "median", "q75", "max", "density_x", "density_x", "densi...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "numeric", ...
$ estimate_value <chr> "8", "8", "-468", "2467", "10060", "11326", "30981", "-433.945", "-373.688982387476", "-...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```


tableCohortTiming

You can also get a table with `tableCohortTiming()`.

```
tableCohortTiming(result = result,
                  timeScale = "years",
                  uniqueCombinations = FALSE)
```

Cohort name reference	Cohort name comparator	Variable name	Estimate name	Estimate value
GiBleed				
anemia	clopidogrel	number records	N	8
		number subjects	N	8
		years_between_cohort_entries	Median [Q25 - Q75]	27.54 [6.75 - 31.01]
			Range	-1.28 to 84.82
	heparin	number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	-11.87 [-11.87 - -11.87]
			Range	-11.87 to -11.87
	osteoarthritis	number records	N	97
		number subjects	N	97
		years_between_cohort_entries	Median [Q25 - Q75]	6.04 [-13.46 - 18.29]
			Range	-48.50 to 42.52
	pneumonia	number records	N	7
		number subjects	N	7
		years_between_cohort_entries	Median [Q25 - Q75]	29.90 [15.57 - 38.99]
			Range	1.77 to 53.45
	rheumatoid_arthritis	number records	N	1

Cohort name reference	Cohort name comparator	Variable name	Estimate name	Estimate value
clopidogrel		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	-17.99 [-17.99 - -17.99]
			Range	-17.99 to -17.99
	anemia	number records	N	8
		number subjects	N	8
		years_between_cohort_entries	Median [Q25 - Q75]	-27.54 [-31.01 - -6.75]
			Range	-84.82 to 1.28
	heparin	number records	N	4
		number subjects	N	4
		years_between_cohort_entries	Median [Q25 - Q75]	-22.87 [-34.14 - -13.19]
			Range	-49.92 to -2.18
	osteoarthritis	number records	N	310
		number subjects	N	310
		years_between_cohort_entries	Median [Q25 - Q75]	-20.17 [-34.93 - -8.85]
			Range	-74.49 to 19.33
	pneumonia	number records	N	7
		number subjects	N	7
		years_between_cohort_entries	Median [Q25 - Q75]	3.11 [2.91 - 5.98]
			Range	2.72 to 11.92
	rheumatoid_arthritis	number records	N	3
		number subjects	N	3
		years_between_cohort_entries	Median [Q25 - Q75]	-3.06 [-14.70 - 3.20]
			Range	-26.35 to 9.45
heparin	anemia	number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	11.87 [11.87 - 11.87]

Cohort name reference	Cohort name comparator	Variable name	Estimate name	Estimate value
	clopidogrel		Range	11.87 to 11.87
		number records	N	4
		number subjects	N	4
		years_between_cohort_entries	Median [Q25 - Q75]	22.87 [13.19 - 34.14]
	osteoarthritis		Range	2.18 to 49.92
		number records	N	35
		number subjects	N	35
		years_between_cohort_entries	Median [Q25 - Q75]	13.11 [-16.33 - 27.61]
	pneumonia		Range	-36.45 to 39.74
		number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	31.87 [31.87 - 31.87]
	rheumatoid_arthritis		Range	31.87 to 31.87
		number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	-5.00 [-5.00 - -5.00]
osteoarthritis	anemia		Range	-5.00 to -5.00
		number records	N	97
		number subjects	N	97
		years_between_cohort_entries	Median [Q25 - Q75]	-6.04 [-18.29 - 13.46]
	clopidogrel		Range	-42.52 to 48.50
		number records	N	310
		number subjects	N	310
		years_between_cohort_entries	Median [Q25 - Q75]	20.17 [8.85 - 34.93]
	heparin		Range	-19.33 to 74.49
		number records	N	35
		number subjects	N	35
		years_between_cohort_entries	Median [Q25 - Q75]	22.87 [13.19 - 34.14]

Cohort name reference	Cohort name comparator	Variable name	Estimate name	Estimate value
		number subjects	N	35
		years_between_cohort_entries	Median [Q25 - Q75]	-13.11 [-27.61 - 16.33]
			Range	-39.74 to 36.45
	pneumonia	number records	N	52
		number subjects	N	52
		years_between_cohort_entries	Median [Q25 - Q75]	43.49 [38.25 - 50.44]
			Range	16.24 to 61.03
	rheumatoid_arthritis	number records	N	18
		number subjects	N	18
		years_between_cohort_entries	Median [Q25 - Q75]	5.61 [-2.01 - 15.46]
			Range	-18.11 to 34.77
pneumonia	anemia	number records	N	7
		number subjects	N	7
		years_between_cohort_entries	Median [Q25 - Q75]	-29.90 [-38.99 - -15.57]
			Range	-53.45 to -1.77
	clopidogrel	number records	N	7
		number subjects	N	7
		years_between_cohort_entries	Median [Q25 - Q75]	-3.11 [-5.98 - -2.91]
			Range	-11.92 to -2.72
	heparin	number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	-31.87 [-31.87 - -31.87]
			Range	-31.87 to -31.87
	osteoarthritis	number records	N	52
		number subjects	N	52
		years_between_cohort_entries	Median [Q25 - Q75]	-43.49 [-50.44 - -38.25]

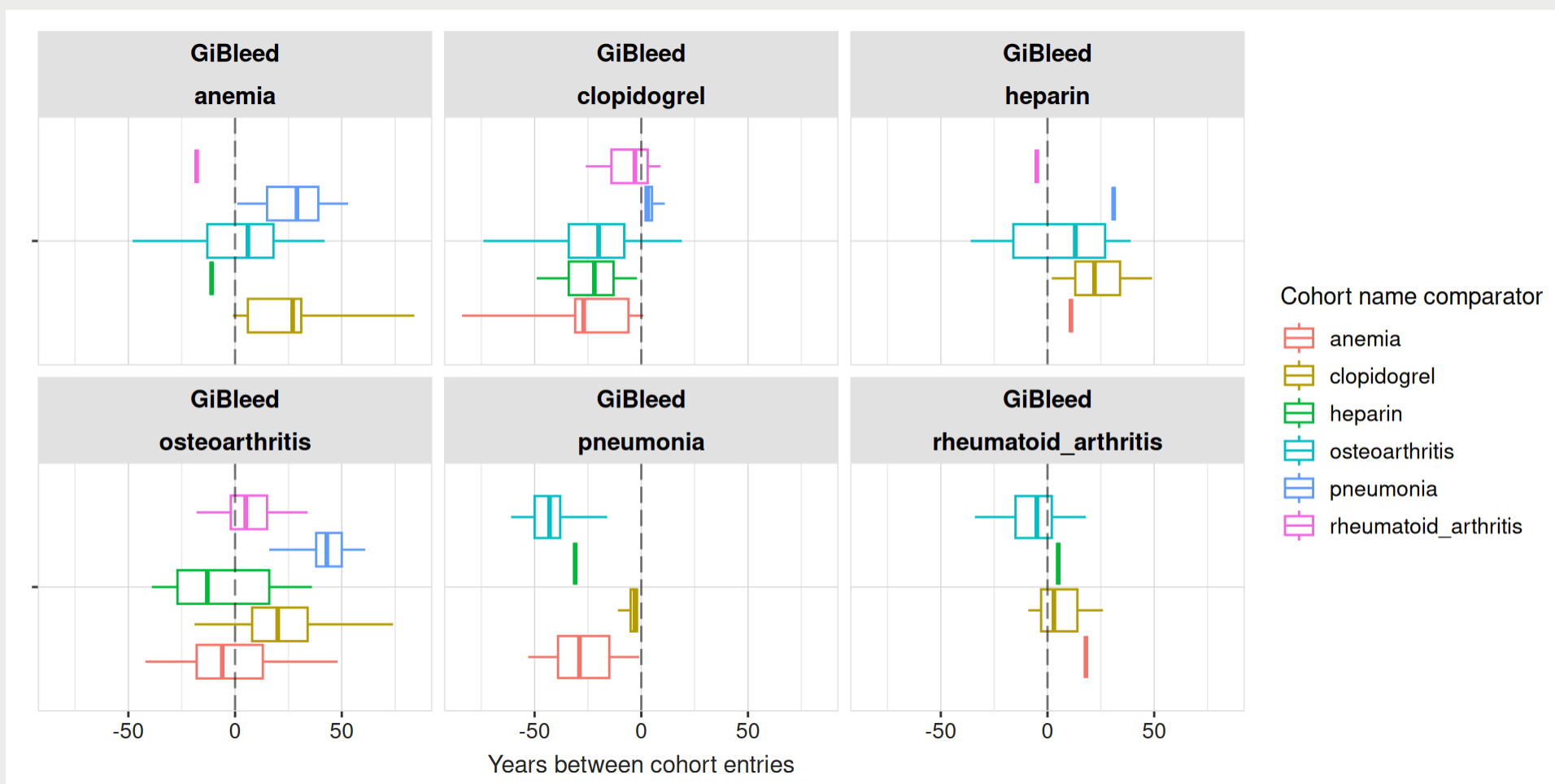
Cohort name reference	Cohort name comparator	Variable name	Estimate name	Estimate value
			Range	-61.03 to -16.24
rheumatoid_arthritis	anemia	number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	17.99 [17.99 - 17.99]
			Range	17.99 to 17.99
	clopidogrel	number records	N	3
		number subjects	N	3
		years_between_cohort_entries	Median [Q25 - Q75]	3.06 [-3.20 - 14.70]
			Range	-9.45 to 26.35
	heparin	number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	5.00 [5.00 - 5.00]
			Range	5.00 to 5.00
	osteoarthritis	number records	N	18
		number subjects	N	18
		years_between_cohort_entries	Median [Q25 - Q75]	-5.61 [-15.46 - 2.01]
			Range	-34.77 to 18.11

[← Back to Presentations](#)

plotCohortTiming

You can also get a plot with `plotCohortTiming()`.

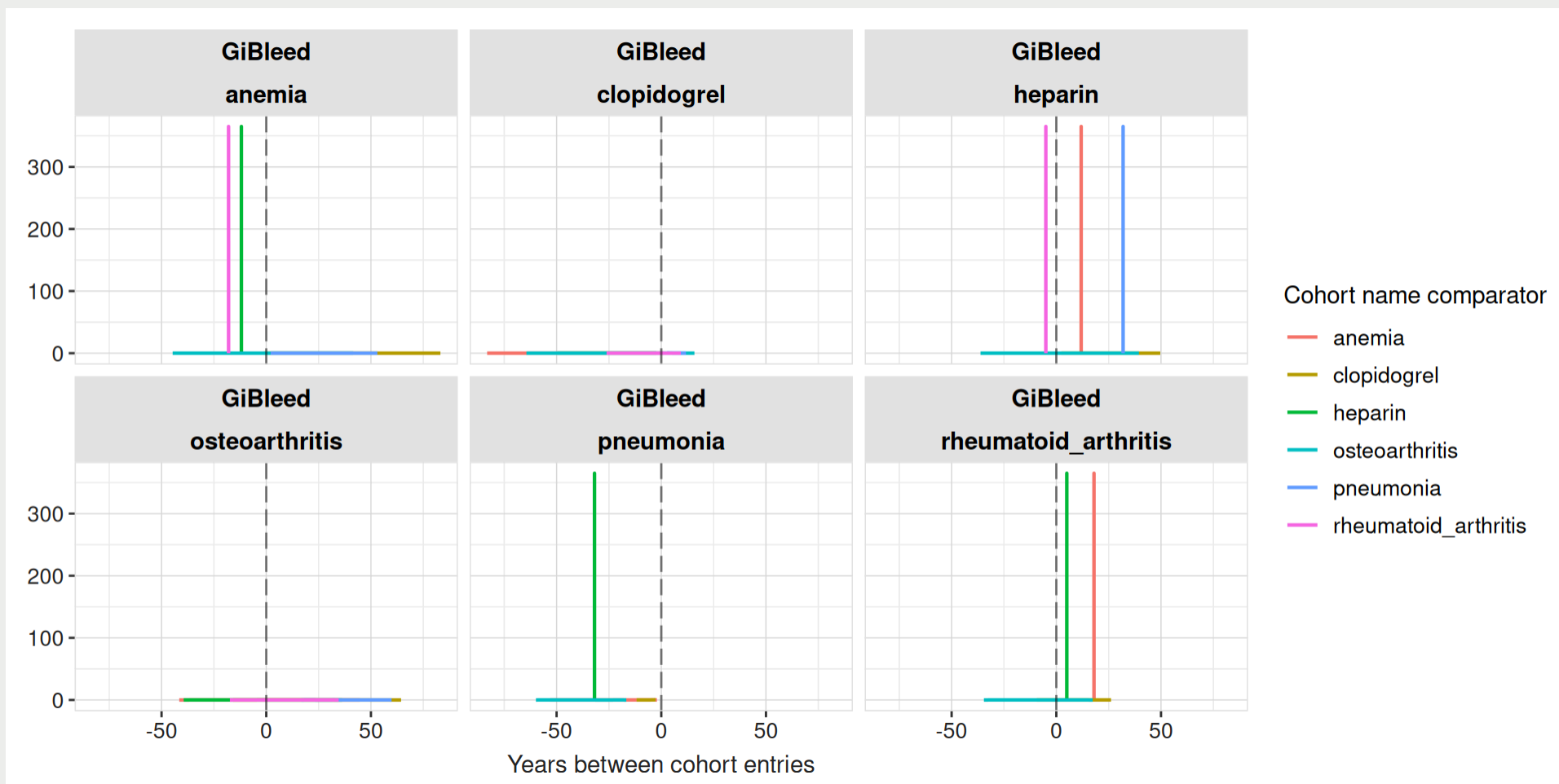
```
plotCohortTiming(  
  result = result,  
  plotType = "boxplot",  
  timeScale = "years",  
  uniqueCombinations = FALSE  
)
```



plotCohortTiming

Or return a density plot.

```
plotCohortTiming(  
  result = result,  
  plotType = "densityplot",  
  timeScale = "years",  
  uniqueCombinations = FALSE  
)
```



Your turn

Can you get the large-scale characteristics from the **drug_exposure** table for the **clopidogrel** cohort table using an anytime-prior time window (excluding index date)? Visualise the result showing the top 5 most frequent codes.

►  [Click to see solution](#)

Top	Estimate value
1	Influenza, split virus, trivalent, injectable, preservative free (40213154) 325 (53.5%)
2	cilostazol 100 MG Oral Tablet (1350311) 46 (7.6%)
3	ibuprofen (1177480) 42 (6.9%)
4	pentoxifylline (1331247) 42 (6.9%)

CohortCharacteristics

👉 Package website

👉 CRAN link

👉 Manual



marta.alcaldeherraiz@ndorms.ox.ac.uk

